Sandbox for "The output gap does not exist"

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1 The single-gap issue

All conventional empirical methods for output gap estimation can be written on the following state space form:

$$\mathbf{s}_t = \mathbf{\Phi}\mathbf{s}_{t-1} + \boldsymbol{\xi}_t, \qquad \boldsymbol{\xi}_t \sim \mathcal{N}(0, \mathbf{\Sigma}_\xi) \tag{1}$$

$$\mathbf{x}_t = \mathbf{H}\mathbf{s}_t + \boldsymbol{\epsilon}_t, \qquad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{\Sigma}_\epsilon). \tag{2}$$

where $\mathbf{s}_t \in \mathbb{R}^{n_s}$ is a vector of unobserved states, $\mathbf{x}_t \in \mathbb{R}^{n_x}$ is a vector of observed variables, $\mathbf{\Phi} \in \mathbb{R}^{n_s \times n_s}$ governs state dynamics, and $\mathbf{H} \in \mathbb{R}^{n_x \times n_s}$ maps states to observables. $\mathbf{\Sigma}_\xi$ and $\mathbf{\Sigma}_\epsilon$ are covariance matrices corresponding to the state innovations, $\boldsymbol{\xi}_t$, and measurement errors, $\boldsymbol{\epsilon}_t$.

The state-space formulation in Equation (1)-(2) includes, among many popular choices for output gap measurement, the HP-filter, band-pass filters, Buterworth filter, Beveridge-Nelson decompositions, Blanchard-Quah decompositions, (multivariate) unobserved components models (Clark, 1987; Kuttner, 1994; Hasenzagl *et al.*, 2022) Laubach-Williams etc, production function approaches, structural VARs, ...

For example, (SOME EXAMPLE HERE, e.g. simple model from Hasenzagl *et al.* (2022) or maybe just AR(2) Harvey-model or Harvey, Trimbur and Van Dijk (2007) model or Watson model?) Vælg én til at starte med.

Which gap does this represent? Common interpretations include inflation-predicting output gap, inflation-relevant output gap, broad business cycle, frequency-restricted fluctuations, ... These labels reflect pragmatic choices - which variables enter $\mathbf{x}_t$, lag structure in $\mathbf{\Phi}$, identifying restrictions — but lack theoretical grounding. In the EXAMPLE ABOVE, the cycle component can for example arguably be interpreted as XXX; A univariate HP filter yields "frequency-restricted fluctuations"; adding inflation to $\mathbf{x}_t$ could be interpreted as an "inflation-signaling gap"

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(Kuttner, 1994); adding financial variables could yield a "credit-augmented gap." The estimated c_t varies with specification, yet is referred to as *the* output gap.

This interpretational flexibility and linguistic convention masks a conceptual problem: economic theory predicts multiple distinct gaps arising from different frictions (nominal rigidities, credit constraints, search frictions) and imperfections (monopolistic market power, ...), each with separate dynamics and policy implications. The single-gap specification embedded in conventional practice creates an identification fallacy.

Fact 1 formalizes the single-gap specification embedded in conventional practice.

Fact 1 (Single-gap assumption in conventional methods)**.** *All conventional methods for output gap measurement restrict the set of possible output gaps to one dimension.*

Sketch "proof" of Fact 1. Let $\mathbf{h}_y$ denote the row of $\mathbf{H}$ corresponding to output. Standard implementations decompose output according to the identity $y_t = \tau_t + c_t$ where τ_t is trend and c_t is cycle, both unobserved components within $\mathbf{s}_t$. The measurement equation for output can thus be written $y_t = \mathbf{h}'_{y,\tau}\mathbf{s}_t + \mathbf{h}'_{y,c}\mathbf{s}_t + \varepsilon_{y,t}$ where $\mathbf{h}_{y,\tau}, \mathbf{h}_{y,c} \in \mathbb{R}^{n_s}$ are selection vectors extracting trend and cycle components from the state vector.

The cycle selection vector $\mathbf{h}_{y,c}$ has exactly one nonzero entry and hence

$$\mathcal{C} = \text{span}\{\mathbf{h}_{y,c}\} \subset \mathbb{R}^{n_s}, \quad \dim(\mathcal{C}) = 1. \tag{3}$$

$\square$

The implication of Fact 1 is stark: by restricting the cycle space $\mathcal{C}$ — the set of possible contemporaneous cycle contributions to output fluctuations — to one dimension, all cyclical variation in output, regardless of the underlying source, must move along a single scalar dimension.

Economic theory, on the other hand, clearly suggests that different frictions create distinct gaps: Afsnit her om teoretiske gab. Måske ny undersektion? Måske noget á la jeg har tilføjet c_t^{fin} blot for at vise det generelle i pointen, men den kan fjernes hvis vi vil fokusere på basic NK til at starte med:

Distinct gaps from different frictions, e.g.

- c_t^f (Flex-price): sticky-price wedge, responds to interest rate policy
- c_t^e (Efficient): Efficient
- c_t^{fin} (Financial): credit cycle
- ...

Each gap exhibits different persistence, volatility, and frequency content. By imposing $\dim(\mathcal{C}) = 1$, conventional methods collapse these into: $c_t \approx w_1 c_t^f + w_2 c_t^e + w_3 c_t^{fin} + \cdots$, where weights $\{w_i\}$ depend on model specification. Good news 1: by careful model specification we could maybe get $\{w_i\} = 1$ and $\{w_{-i}\} = 0$. Good news 2: nothing restricts us from specifying multiple gaps, as in theory.

References

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